Lab 2 – Rectangular Waveguide

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Pre-Lab

Obj 1

The geometry, frequency range, and dominant mode cutoff frequency were found from the X-Band specification. The dominant mode is TE10 for rectangular waveguide where a>b. The E field has only y component and is strongest at a/2 (center guide's longer side). The guided wavelength and wave impedance of the dominant mode at 10 GHz were calculated using MATLAB.

Obs 1) X-band wave guide $f = 8.20 - 12.4$ GHz, TE_{10} cutoff 6.557 GHz

$a = 2.286$ $b = 1.016$

for TE_{10}

$k_y = \frac{n\pi}{b} = 0$ $k_x = \frac{m\pi}{a} =$ $k_{c,1,0} = \sqrt{k_x^2 + k_y^2} = k_x$

$e_x = [\sim\sim]\cdot k_y = 0$ $e_z = 0$

$E = e_y = \frac{j\omega\mu}{k_x} A_{mn} \sin(x k_x)$ → strongest @ $x = \frac{\pi}{2k_x} = \left(\frac{a}{2}\right)$

Using matlab: $f_{c,mn}^{TE} = \frac{1}{2\sqrt{\mu\epsilon}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$

first two mode cutoffs:

- 6.557 GHz (TE_{10})
- 13.11 GHz (TE_{20}) ∴ only TE_{10} propagates in 8.2 – 12.4 GHz

@ 10 GHz: $Z_{TE10} = 498.97\,\Omega$ $\lambda = 3.97$ cm

Obj 2

The cutoffs for the first few TE and TM modes were found using MATLAB. The first cutoff mode is TE20. This mode's E-field is strongest at x = a/4 or 3a/4. (since Kx2 is doubled). The attenuation was found for a guide with length 4*lambda10.

Obj 2) $Z_{TE20} = \frac{\omega\mu_0}{\beta_{20}} = j440\Omega$ // strongest @ $x = \frac{\pi}{2k_x} = \frac{a}{4}$ or $\frac{3a}{4}$ //

$\beta_{20} = \sqrt{k_0^2 - k_{c20}^2} = -j177.8$

$att_n = 20\log\left(e^{-j4\lambda_{10}\cdot\beta_{20}}\right) = 245.2\,dB$ //

Obj 3

Since the attenuator section has a/2 width, it cuts off the dominant mode TE10. The length is calculated using the attenuation constant found by MATLAB.

Obj-3) $Z_{TE10} = \frac{\omega\mu_0}{\beta_{10}}$ $\beta_{10} = \sqrt{k_{10}^2 - k_{c10}^2}$ $k_{c10} = \sqrt{\left(\frac{m\pi}{a/2}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$

$= j440\Omega$ $= -j177.8$ $\alpha = 177.8$ //

$att_n = 20\log\left(e^{-\alpha l}\right) = 60\,dB$

$l = 3.89\,cm$ //

Lab

Part 1

The model shown in Figure 1 was used for all part 1 simulations. As designed in the pre-lab, it has length 4*lambda, and X-Band dimensions 2.286 cm x 1.016 cm.

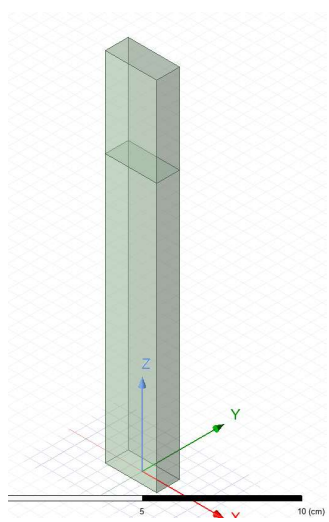


Figure 1: Device model for X-band rectangular waveguide

Wave ports were set at each end as shown in Figures 2 and 3. Renormalization was disabled. Only the dominant mode was simulated.

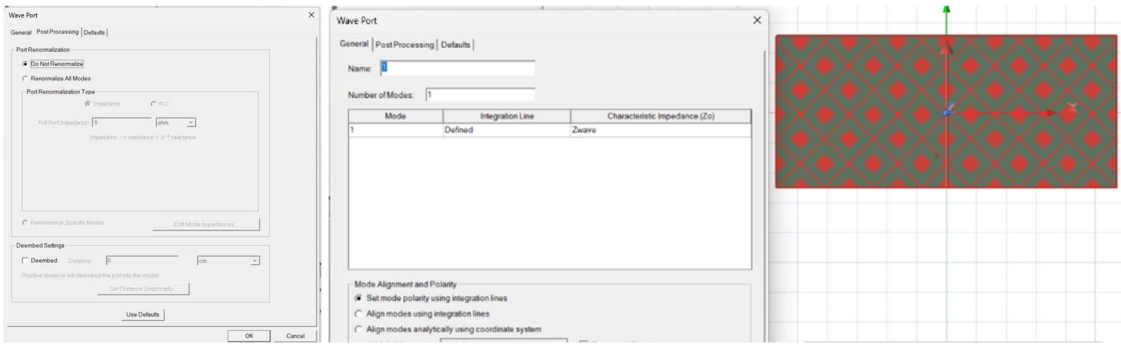


Figure 2: Port 1 setup for X-band rectangular waveguide

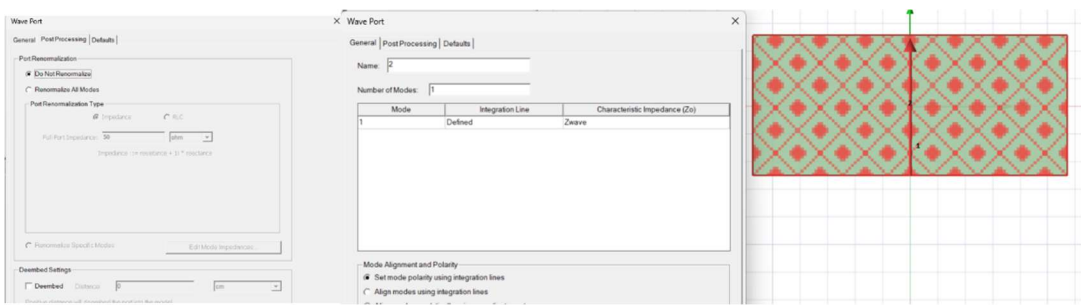


Figure 3: Port 2 setup for X-band rectangular waveguide

(Res-1) Figure 4 shows the plot of S(1,1), which represents the reflection seen at the input port. Figure 5 shows the plot of S(2,1), which represents the attenuation seen at the output port. Reflection remains low and the output ratio remains at 0 dB across the operating band. This matches expectations, since the ports are matched and there is no attenuation for the dominant mode.

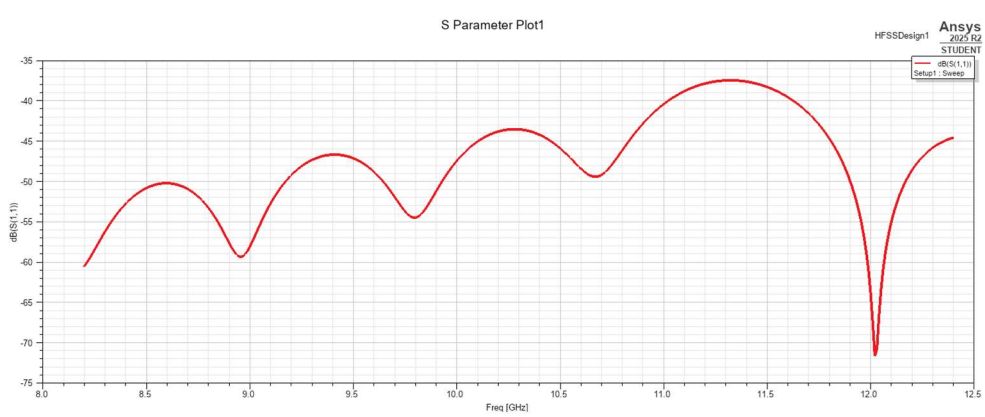


Figure 4: S(1,1) operating band sweep for X-band rectangular waveguide

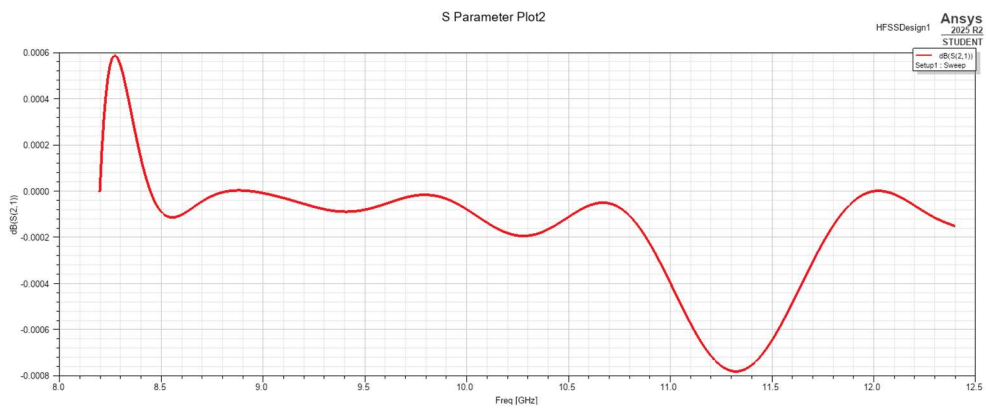


Figure 5: S(2,1) operating band sweep for X-band rectangular waveguide

(Res-2) Shown in Figure 6, the simulated wave impedance closely matches the predicted value.

Freq		S:1	S:2	Port Zo:1
10GHz	1	(0.0042157, -0.00035201)	(0.99961, 0.027685)	(498.98, 0)
	2	(0.99961, 0.027685)	(-0.0041902, -0.0005849)	(498.98, 0)

Figure 6: Simulated wave impedance for X-band waveguide

(Res-3) Shown in Figure 7, the simulated E-field of the dominant mode matches the expectation of being strongest at a/2 and satisfies the boundary condition of being 0 at the edges as expected. The irregular pattern of the field lines can be attributed to simulation quirks (i.e. direction can become unstable when magnitude is very small).

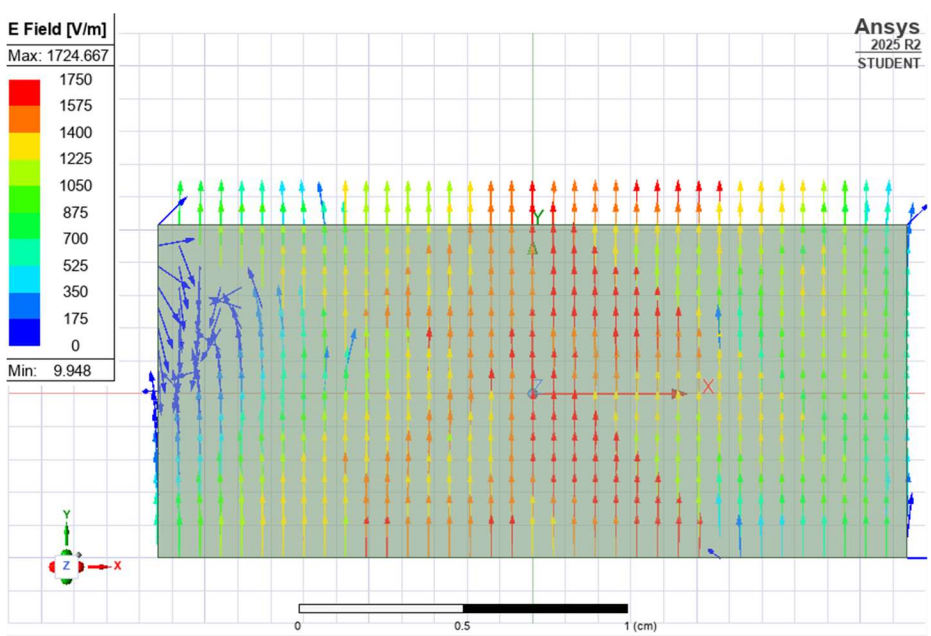


Figure 7: Simulated E-field for dominant mode of X-band waveguide

Renormalized Port-1

Using the same model and port setup, port 1 is renormalized to 50 Ohms, as shown in Figure 8

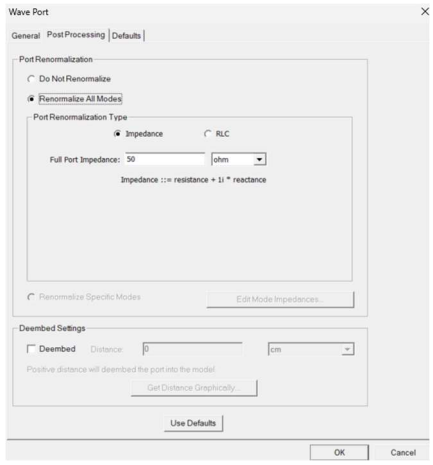


Figure 8: Port-1 renormalization to 50 Ohms

(Res-4) Figures 9 and 10 show the updated scattering parameters after renormalization. The results are clearly different from before, because the terminating impedance is no longer matched, causing reflection to occur at the port.

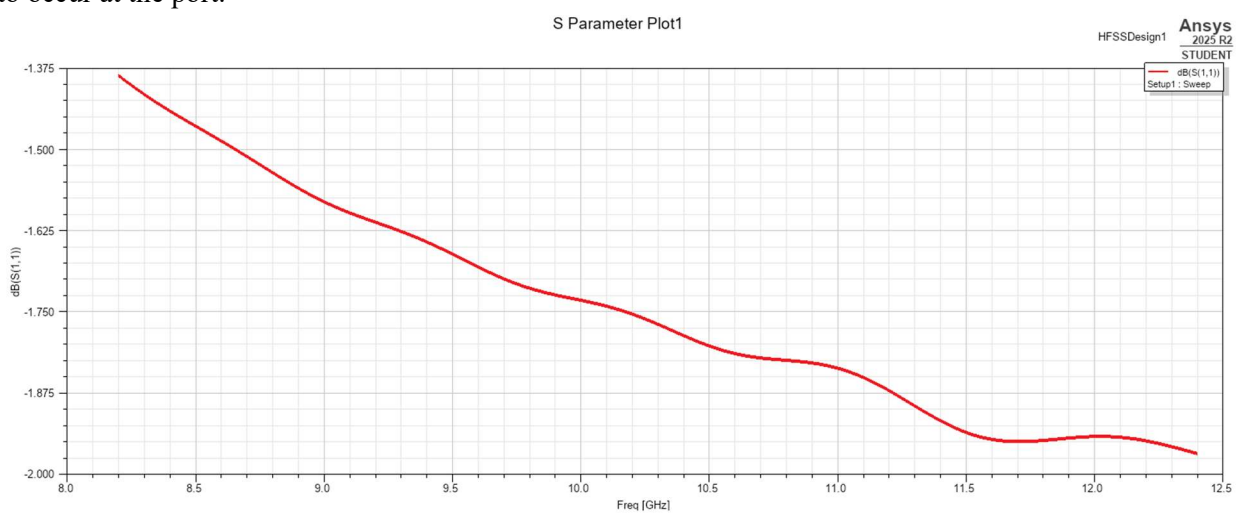


Figure 9: S (1,1) sweep for X-Band waveguide with renormalization

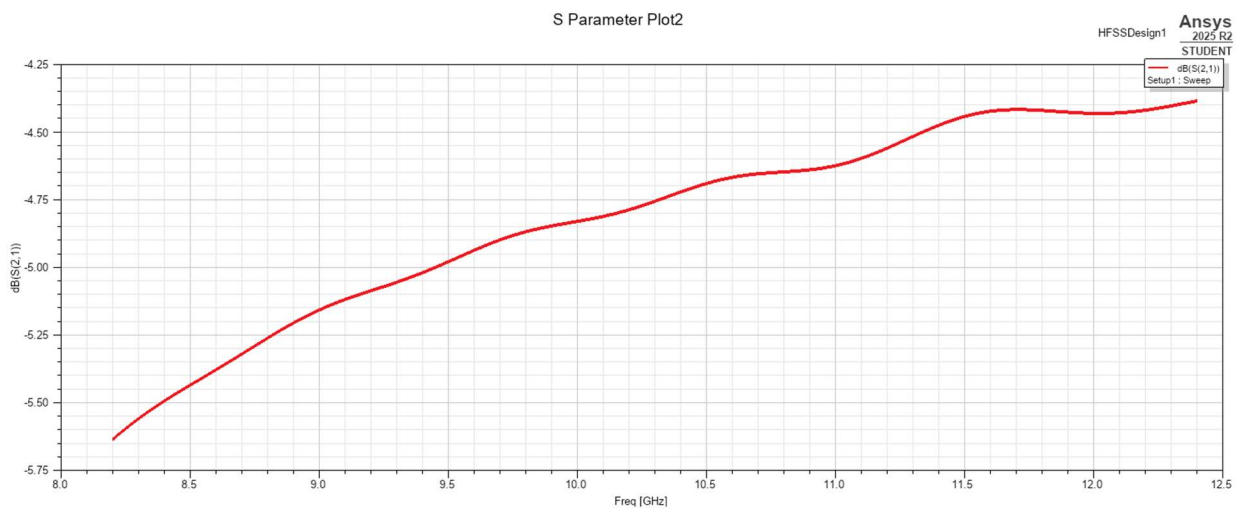


Figure 10: S (2,1) sweep for X-Band waveguide with renormalization

Added Second Wave Mode

Using the original port setup (no renormalization), the ports were modified to include a second mode. For each port, an integration line was added at a = 3/4 which is where the field is strongest for this mode. These changes are shown in Figure 11.

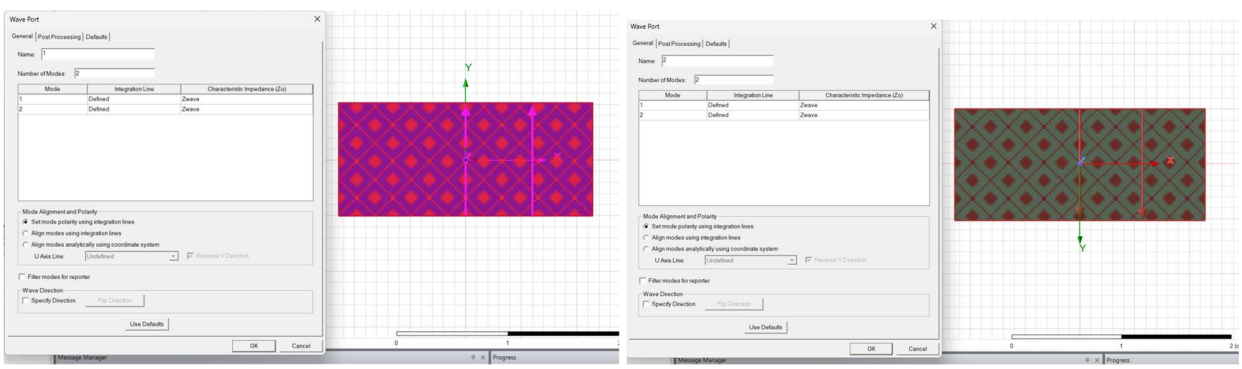


Figure 11: Port setup for second wave mode analysis

(Res-5) The S-parameter plots for each wave mode are shown in Figure 12. As expected, both modes see limited reflection at the input port, since the termination is matched. Furthermore, the second mode is cut off across the operating band as designed.

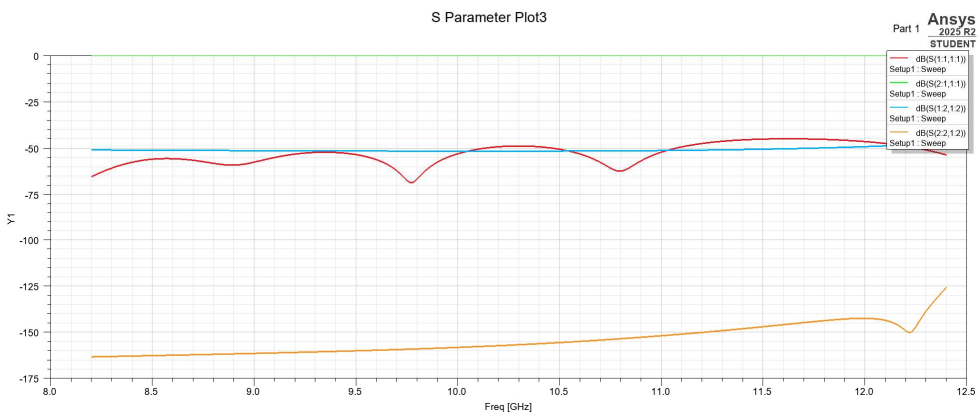
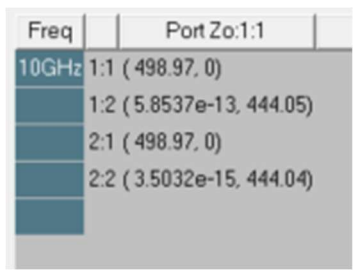


Figure 12: S(1,1) and S(2,1) for first and second wave modes

(Res-6) Figure 13 shows the simulated wave impedances for the first and second modes. The first mode (TE10) wave impedance perfectly matches the expected value. The second mode (TE20) wave impedance is also close to the expected value of 440 Ohms.



Freq	Port Zo:1:1
10GHz	1:1 (498.97, 0)
	1:2 (5.8537e-13, 444.05)
	2:1 (498.97, 0)
	2:2 (3.5032e-15, 444.04)

Figure 13: Wave impedances for first and second wave modes

Part 2

The model shown in Figure 14 was used for all part 2 simulations. The two X-band sections on the sides are each one guided wavelength long, while the cutoff section is 3.89 cm as designed in the prelab, with a width of a/2

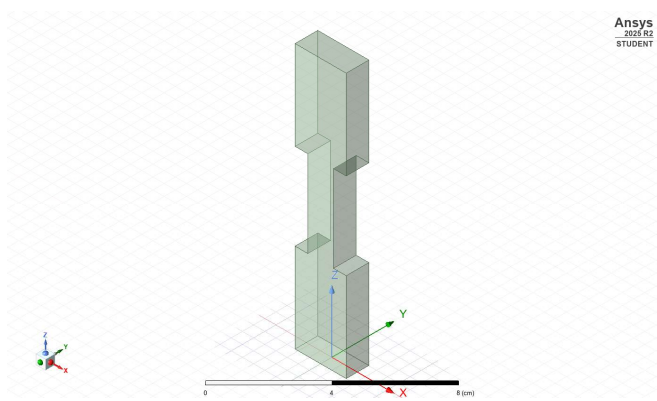


Figure 14: Device model for X-band attenuator

Wave port setups were identical to those in part 1.

(Res-7) Renormalization was disabled to find the scattering parameters with zero reflection at the ports. Thus S(1,1) = Gamma(1). Only the dominant mode was simulated.

(Res-8) Figures 15 and 16 show the scattering parameters of the attenuator. The S(1,1) plot shows almost total reflection, which matches expectations. Since the wave is cut off at the narrow section and the system is lossless, power conservation demands that most of the power is reflected to the input port. The S(2,1) plot shows -60.5 dB of attenuation at 10 GHz, which matches the designed -60 dB.

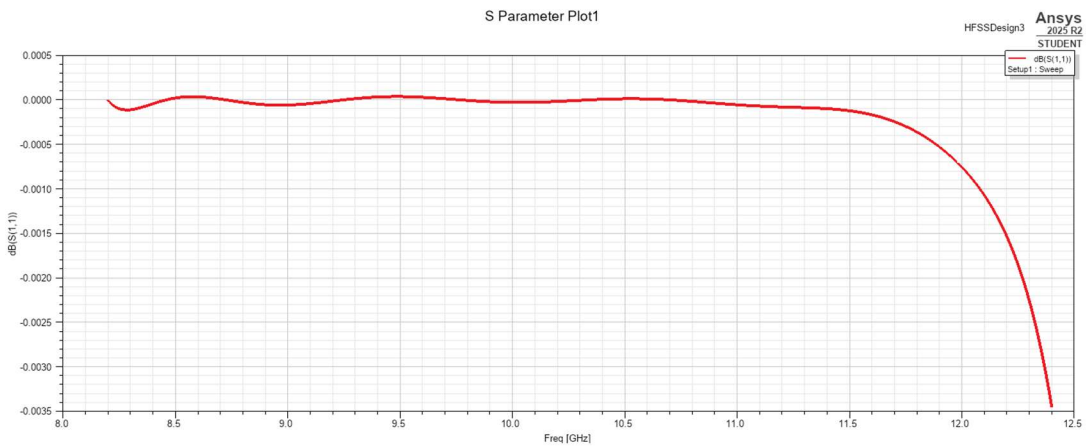


Figure 15: S(1,1) plot for attenuator

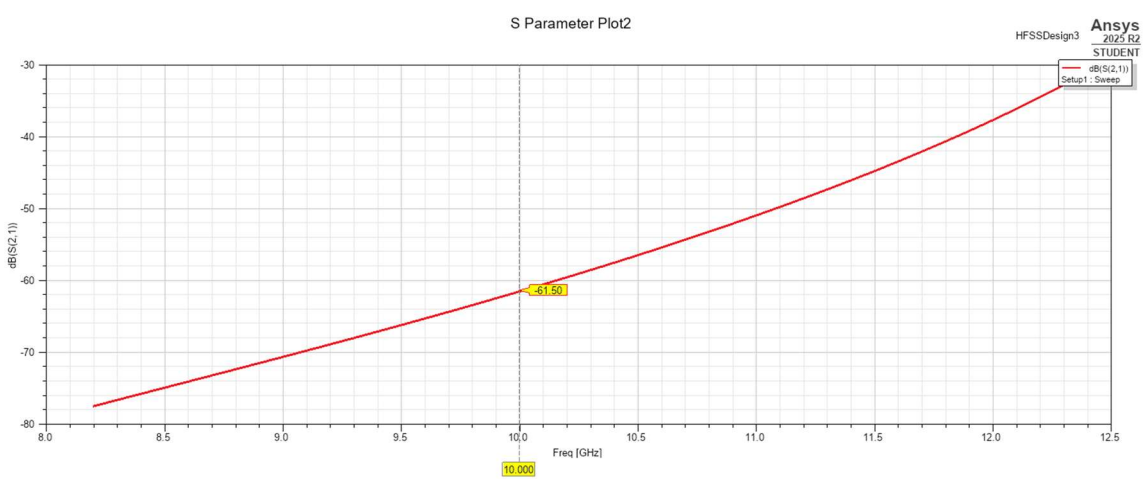


Figure 16: S(2,1) plot showing attenuation at 10 GHz

(Res-9) Figure 17 shows the power flow inside the attenuator. As expected, the field diminishes sharply after entering the cutoff section and remains zero from there onward.

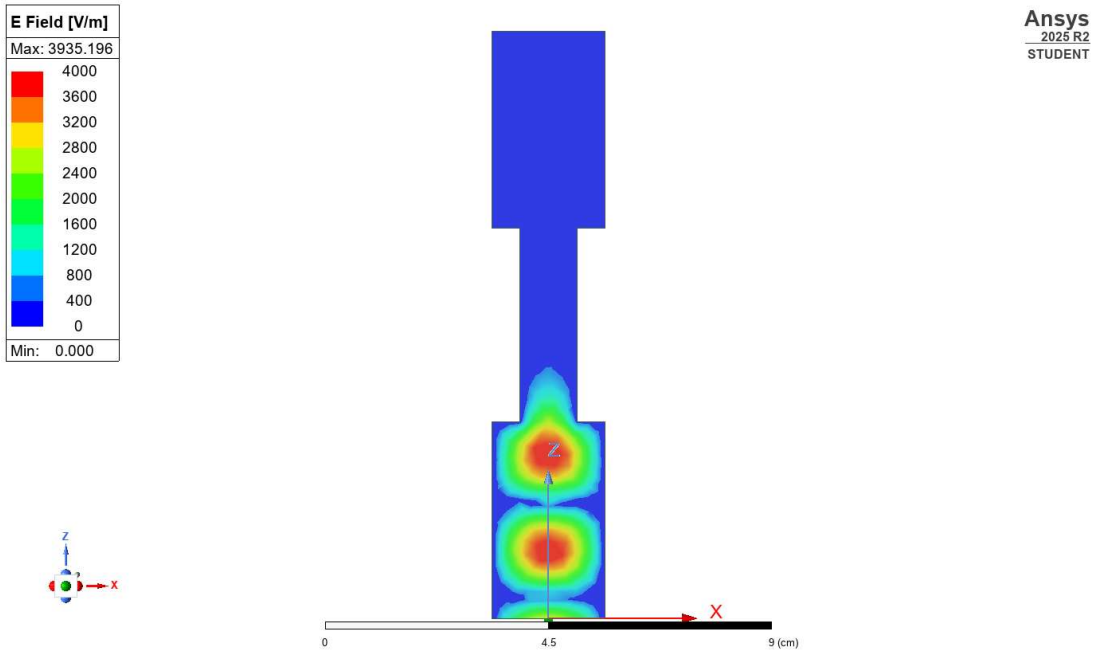


Figure 17: Still frame of power flow animation

Part 3

The model shown in Figure 18 was used for all part 3 simulations. Each X-band section is two guided wavelengths long from port to corner. The design parameters were set up as shown to be adjustable.

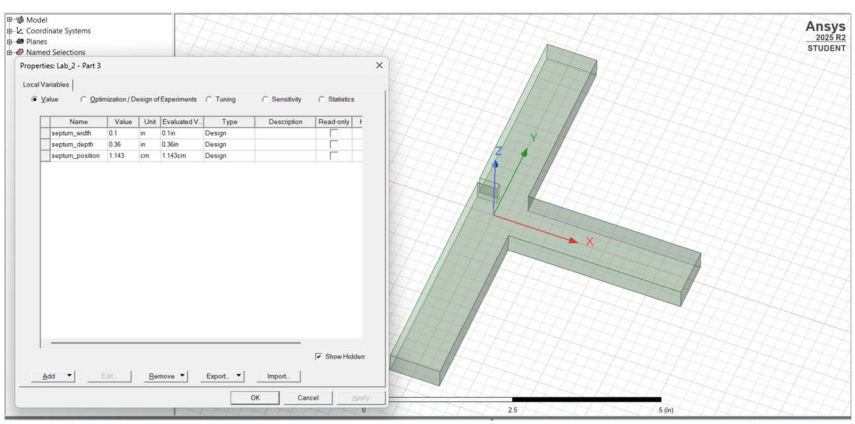


Figure 18: Device model for X-band attenuator

Wave port setups at each port were identical to those in part 1. As with part 2, renormalization was disabled.

(Res-10) The optimal solution for septum depth was found using a parametric sweep. The results are shown in Figure 19. The depth was optimized at 0.36 in.

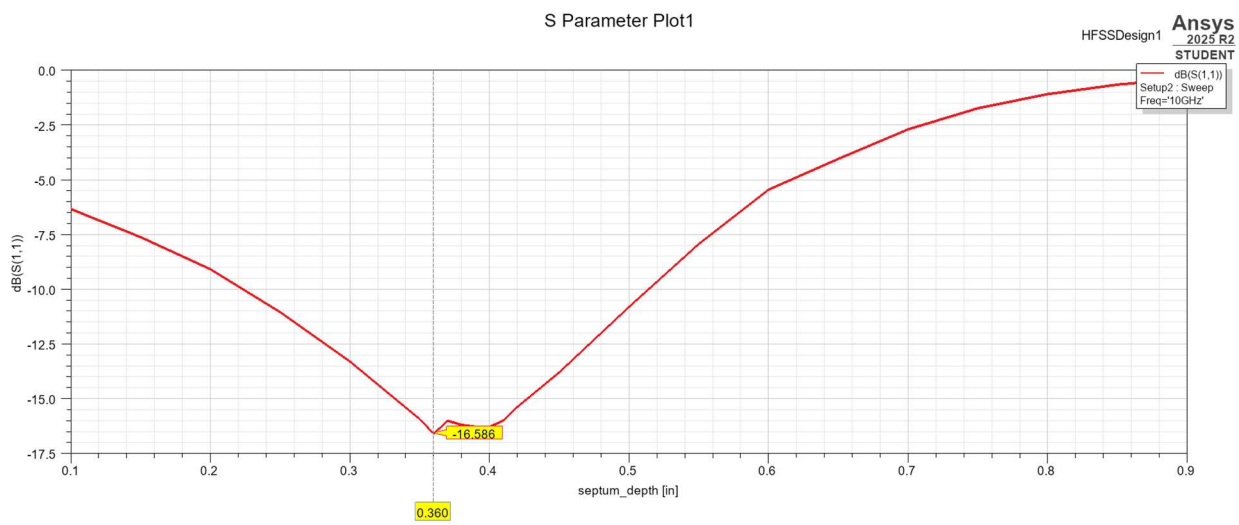


Figure 19: Septum depth parameter sweep

(Res-11) The solution for septum position was found by setting depth to the previously determined value (0.36 in.) and using a parametric sweep to find the point where S(2,1) – S (3,1) = 3 dB. As shown in Figure 20, the solution was found to be 0.4 cm (0.16 in.).

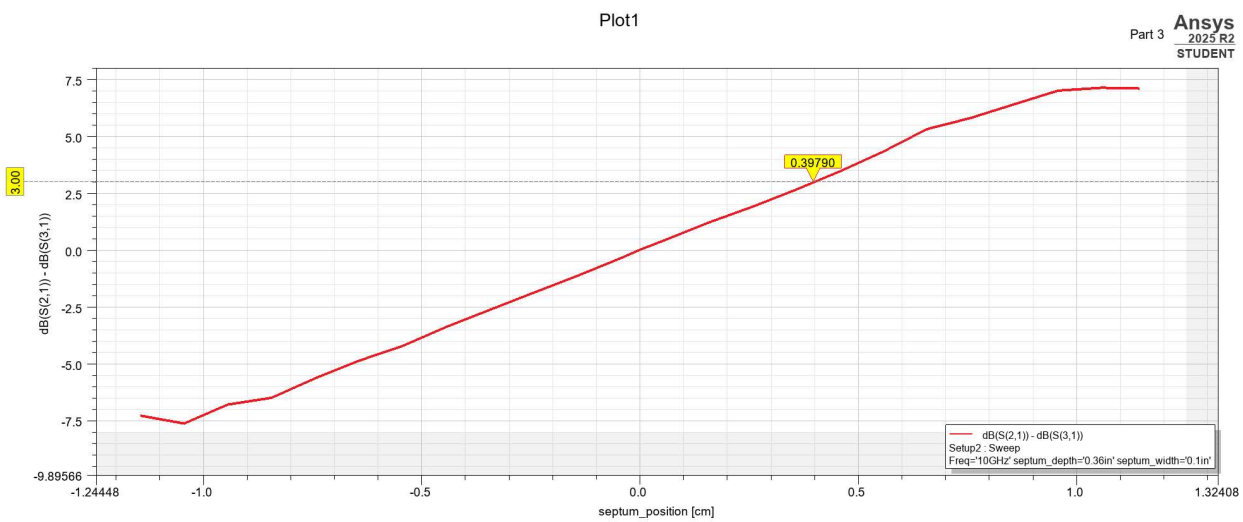


Figure 20: Septum position parameter sweep

(Res-12) The S parameters, shown in Figure 21, clearly show the expected device behavior: most of the power gets split between ports 2 and 3, with the 3 dB difference clearly shown again at 10 GHz. This behavior can also be seen in the power flow animation still shown in Figure 22, with the power of the incident wave from port 1 being split unevenly with a larger portion going towards port 2 (bottom).

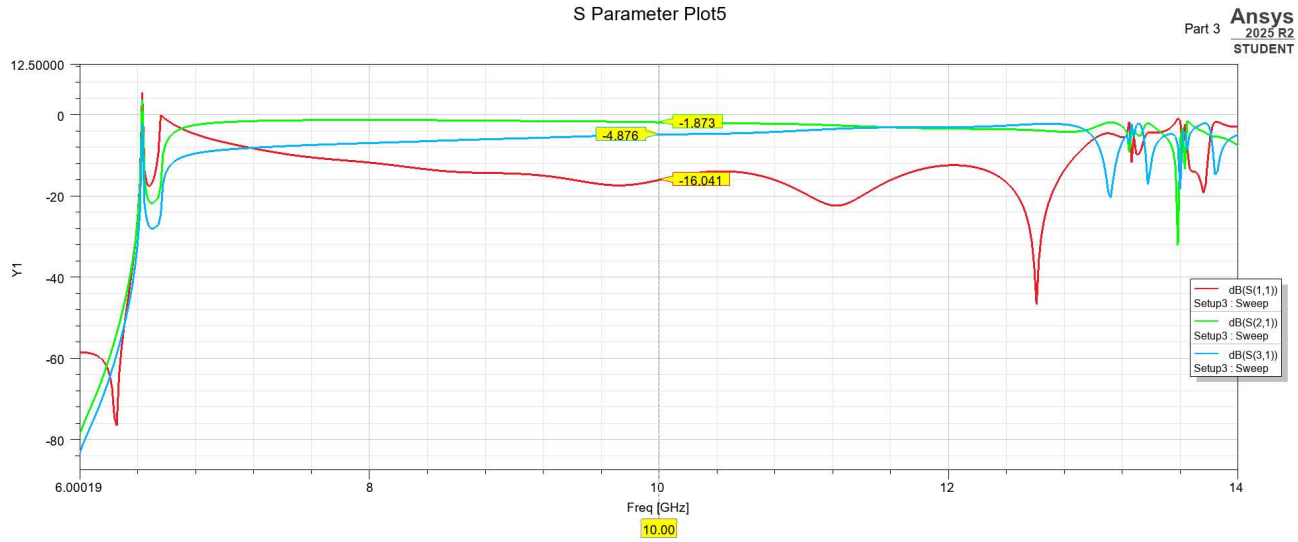


Figure 21: Frequency sweep for designed wave splitter device

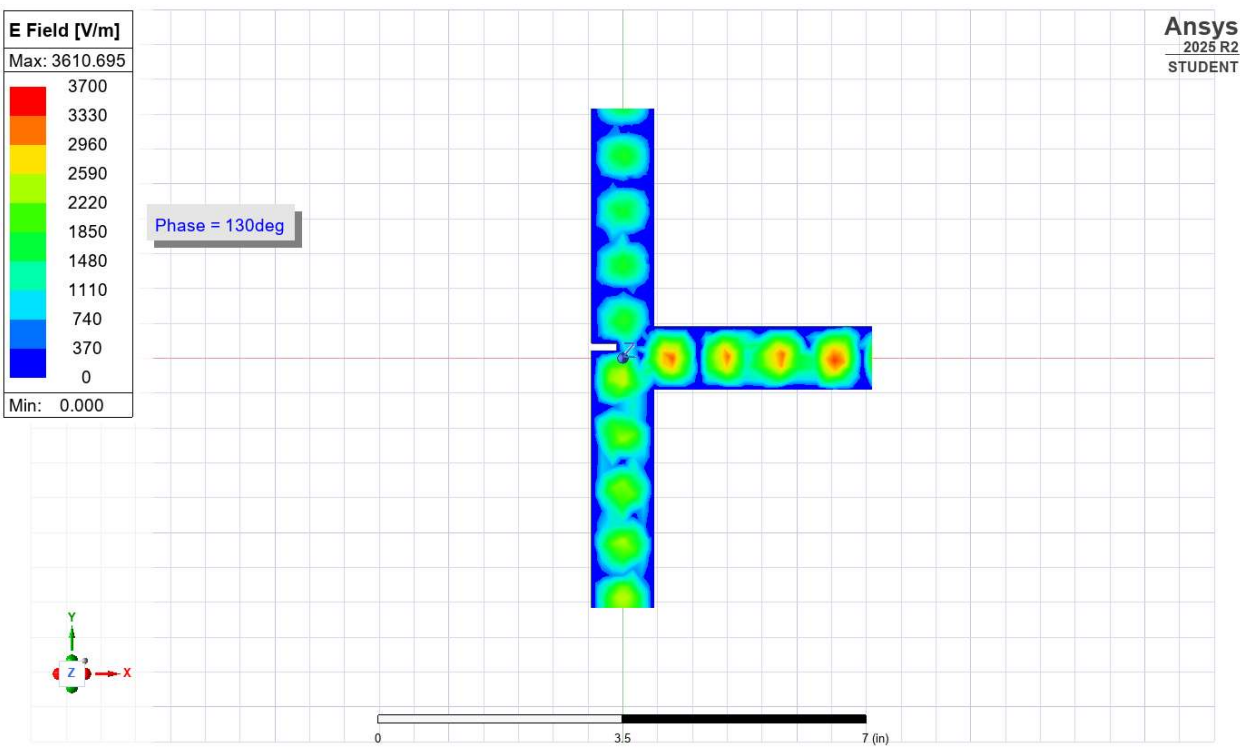


Figure 22: Power flow for wave splitter device

Report Checklist

1. I have included screenshots of all the HFSS models. **AM**
2. I have included screenshots of all port setups, including integration lines where appropriate. **AM**
3. I have included all necessary results in graphs, tables, or statements. **AM**
4. I have interpreted and commented on all my results. **AM**